NEWLANDS' LAW OF OCTAVES

Class 10 · Science Part 1 · Ch-2 Periodic Classification of Elements

SCIENTIST & YEAR

① **John Newlands** – English scientist

② Year : **1866**

③ He correlated the atomic masses of elements to their properties in a different way.

HOW HE ARRANGED THE ELEMENTS

1. Took all the elements known at that time
↓
2. Arranged them in **increasing order of atomic mass** (बढ़ते क्रम में)
↓
3. Started with the lightest element **Hydrogen (H)**
↓
4. Ended up with **Thorium (Th)**
↓
5. Observed : **every 8th element** had properties similar to the 1st.

* **WORD MEANING**
Octave (अष्टक) – a group of **eight**; in music, the 8th note which repeats the 1st note.

RECALL :- WHY CLASSIFY ELEMENTS ?

① Today **118 elements** are known to the scientific world.

② Around the year **1800** only about **30 elements** were known.

③ More elements were discovered in the course of time.

④ ∴ To **ease the study** of such a large number of elements, scientists started searching for a **pattern**.

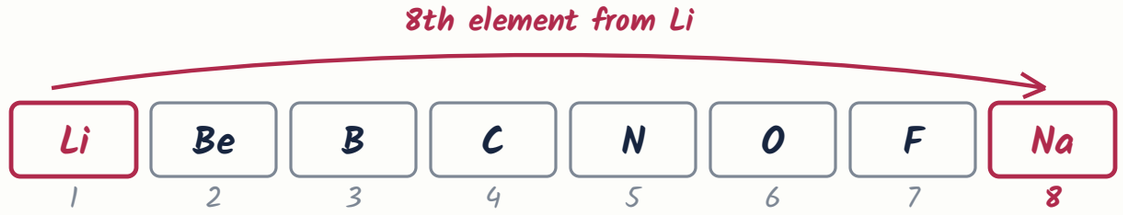
DEF^{N} :- LAW OF OCTAVES

① When the elements are arranged in the increasing order of their atomic masses,

② the properties of **every eighth element** are similar to those of the **first element**.

③ This similarity observed b/w the 8th & the 1st element is called the **Law of Octaves**

Example :- Li → Na



∴ Li and Na show similar properties

More Examples

① **Mg shows similarity to Be**

② **Cl shows similarity with F**

1ST ELEMENT	8TH ELEMENT	NATURE
Li (6.9)	Na (23.0)	Both metals
Be (9.0)	Mg (24.3)	Both metals
F (19)	Cl (35.5)	Both nonmetals

(Atomic masses as given in the text book)

RK^{N}
Newlands **compared this similarity with the octaves in music** – that is why the law got the name **'Law of Octaves'**.

WHY THE NAME 'OCTAVE' ?

RK^N
*The word **'octave'** is borrowed from **music**, not from chemistry. The text book explains it in the **'Do you know ?'** box on page 17.*

A) Indian Music System – Saptak

1. There are **seven main notes** :
   Sa, Re, Ga, Ma, Pa, Dha, Ni
2. Their collection is called **Saptak** (सप्तक)
3. The **frequency** (आवृत्ति) of the notes goes on increasing from 'Sa' to 'Ni'.
4. Then comes the 'Sa' of the **upper Saptak**, at **double the frequency** of the original 'Sa'.
5. ∴ The notes **repeat** after completion of one Saptak.

B) Western Music System – Octave

1. The seven notes are :
   Do, Re, Mi, Fa, So, La, Ti
2. The note **'Do'** having **double the original frequency** comes again at the **eighth place**.
3. This is called the **Octave** of western notes.
4. Music is created by the **variety** in the use of these notes.

KEY POINT FOR THE BOARD
7 notes + repeat of the 1st note = 8 → that 8th place is the **octave**.

* **WORD MEANING**
Frequency (आवृत्ति) – number of vibrations per second; higher frequency = higher pitch.

The Octave – Picture

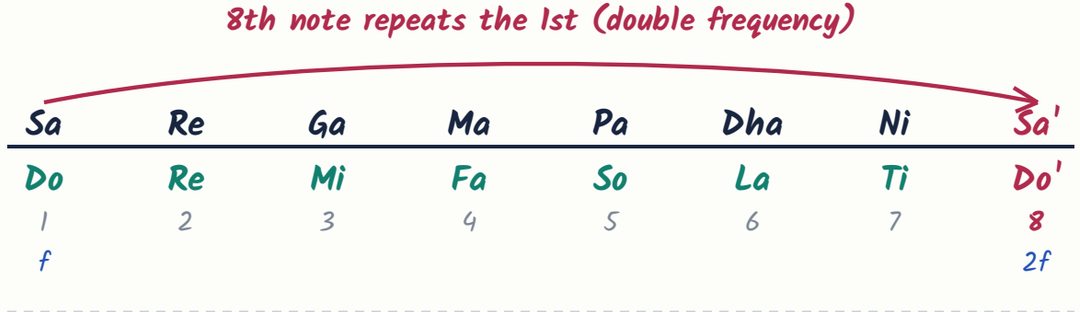


Fig. – Octave in music and in elements

Comparison

IN MUSIC	IN NEWLANDS' ARRANGEMENT
Notes arranged in increasing **frequency**	Elements arranged in increasing **atomic mass**
8th note = **same** as the 1st note	8th element = properties **similar** to the 1st
Sa → Sa' / Do → Do'	**Li → Na , Be → Mg , F → Cl**

EXAM POINT
Q. Why is Newlands' law called the **'Law of Octaves'** ?
A. Because, just as the **8th note** of a musical octave repeats the 1st note, the **8th element** in Newlands' arrangement has properties **similar** to the 1st element.

RK^N
Only the **name** is taken from music. The **basis** of Newlands' arrangement is always the **atomic mass**.

NEWLANDS' OCTAVES – THE TABLE

DO (SA)	RE (RE)	MI (GA)	FA (MA)	SO (PA)	LA (DHA)	TI (NI)
H	Li	Be	B	C	N	O
F	Na	Mg	Al	Si	P	S
Cl	K	Ca	Cr	Ti	Mn	Fe
Co & Ni	Cu	Zn	Y	In	As	Se
Br	Rb	Sr	Ce & La	Zr	–	–

Fig. 2.2 – Newlands' Octaves

Size Of The Table

① Newlands fitted all the known elements in a table of
7 × 8 = 56 boxes

② 7 vertical columns = 7 musical notes.

③ In some boxes he had to place **two elements** together. **e.g.:- Co & Ni , Ce & La**

HOW TO READ THE TABLE
Elements written in the **same vertical column** (under the same note) were expected to show **similar properties**.

While Drawing On The Board

1. Draw 7 columns – write the note names on top.
↓
2. Fill row by row, **left → right**, in increasing atomic mass.
↓
3. Circle Co & Ni and Ce & La – the two double boxes.

Reading The Columns

① Under **'Re'** → **Li, Na, K, Cu, Rb** – all alkali metals, correctly grouped.

② Under **'Do'** → **H, F, Cl, Br** – halogens correctly grouped... but Co & Ni also sit here.

③ Under **'Mi'** → **Be, Mg, Ca, Zn, Sr**

→ LINK WITH DOBEREINER'S TRIADS (SAME CH-2)

POINT	DOBEREINER	NEWLANDS
Year	1817	1866
Country	German	English
Group size	Group of **3** (Triad)	Every **8th** element (Octave)
Basis	Atomic mass	Atomic mass
Rule	At. mass of middle element = mean of other two	8th element resembles the 1st
Main defect	All known elements could not be fitted into triads	Valid only up to calcium (Ca)

ORDER OF THE THREE ATTEMPTS
Dobereiner (1817) → Newlands (1866) → Mendeleev (1869–1872) → Modern periodic table (Moseley, 1913).

RKN
Both Dobereiner and Newlands used **atomic mass** as the basis. The same basis was later used by **Mendeleev** for his periodic table.

LIMITATIONS OF NEWLANDS' OCTAVES

1. The law was found to be applicable **only up to calcium (Ca)**. Beyond Ca the 8th element did not resemble the 1st.
2. The table had only **56 boxes**, so Newlands had to place **two elements in some boxes**.
   e.g.:- Co & Ni , Ce & La
3. Elements with **different properties** came under the **same note**.
   e.g.:- the metals Co and Ni were placed under the note 'Do' along with the halogens.
4. **Fe**, which is similar to Co & Ni, was placed **away** from them under the note 'Ti', along with the nonmetals **O** and **S**.
5. The octaves had **no provision** (व्यवस्था नहीं) to accommodate the **newly discovered elements**.
6. Properties of the elements discovered later **did not fit** in the law of octaves.

BOARD TRICK TO REMEMBER (5 DEFECTS)
Calcium limit → **2** in a box → **wrong** company → **Fe** misplaced → **no space** for new elements.

Conclusion

1. Because of these defects, Newlands' octaves **could not be accepted** as a complete classification.
2. But it was an **important step** – it was the first attempt to arrange **all** the known elements in **one** table.
3. The next and the most important step was taken by **Mendeleev** (1869–1872).

Defect Shown In The Table

NOTE	ELEMENTS PLACED TOGETHER	WHY WRONG ?
Do	F, Cl, Br + Co & Ni	Halogens (nonmetals) with **metals**
Ti	O, S, Fe, Se	**Fe** (metal) kept with nonmetals, far from Co & Ni

TEXT BOOK EXERCISE – ANSWER
Column 1 : **Octave** → Column 2 : **d.** Properties of the eighth element similar to the first → Column 3 : **3. Newlands**

REVISION – ONE LINE EACH

1. **Scientist :** John Newlands, English, year 1866.
2. **Basis :** increasing order of atomic mass, H → Th.
3. **Law :** every 8th element has properties similar to the 1st.
4. **Name :** taken from the octave of music (Sa...Ni → Sa' / Do...Ti → Do').
5. **Table :** 7 × 8 = 56 boxes, Fig. 2.2.
6. **Defects :** valid only up to Ca, 2 elements in one box, unlike elements together, Fe misplaced, no room for new elements.

HOME WORK
Write the Law of Octaves with two examples, draw Fig. 2.2 from memory, and write any **four** limitations of Newlands' octaves. (Text book Page 17)